

Topic B – Bayesian Decision and Uncertainty: Skin Lesion Triage

Dataset	HAM10000: dermoscopy images with diagnostic labels. Related challenge: ISIC 2018.
Core task	Binary melanoma-vs-benign triage with calibrated uncertainty.
Main question	How should a classifier behave when false negatives are much more costly than false positives?

Required work

- Classical pipeline: HSV colour histograms and a Gaussian mixture model per class.
- Decision rule: Bayesian decision theory with asymmetric loss.
- Modern pipeline: frozen EfficientNet-B0 with a trained classification head.
- Uncertainty: use Monte Carlo Dropout or an equivalent uncertainty estimate.

Expected outputs: ROC curves, reliability or calibration analysis, uncertainty examples, and a comparison of classical and deep triage decisions.

Theory focus: Bayes decision boundaries, asymmetric costs, calibration, and the meaning of MC Dropout.